

ProcureSense

AI-Powered Procurement Contract Intelligence Assistant

Project ID	PS294
Domain	HR & Corporate Functions
Sub-Domain / Process	Contract Procurement
AI Focus	LLM + Retrieval-Augmented Generation (RAG)
Business Function	Procurement
Potential Data Inputs	Vendor Contracts, PO History, Procurement Playbooks
Expected Deliverable	Contract Obligation Dashboard

Abstract

Procurement contracts contain critical information such as renewal dates, pricing clauses, service obligations, and compliance requirements. Manual contract review is time-consuming and often leads to missed renewals, cost escalations, and operational risks.

This project proposes **ProcureSense**, an AI-powered contract intelligence assistant built using NVIDIA Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG). The system automatically analyzes procurement contracts, extracts key contractual information, and enables natural language interaction with contract documents. A centralized Contract Obligation Dashboard provides visibility into renewals, obligations, vendor commitments, and potential risks, helping procurement teams improve efficiency and decision-making.

1. Problem Statement

Procurement contracts are often stored as lengthy and unstructured documents. Identifying important details such as renewal dates, escalation clauses, payment terms, and service obligations requires extensive manual effort. This can result in missed deadlines, compliance risks, delayed decisions, and reduced visibility into contractual commitments.

2. Objectives

1. Develop an AI-powered procurement contract assistant.
2. Automatically identify renewal dates and escalation clauses.
3. Extract vendor obligations, payment terms, and compliance requirements.
4. Enable natural language querying of contract documents.
5. Generate alerts for renewals and contractual risks.
6. Build a Contract Obligation Dashboard for monitoring and analysis.

3. Proposed Solution

The proposed system utilizes NVIDIA's Generative AI ecosystem and a Retrieval-Augmented Generation architecture. Contract documents are uploaded, processed, converted into vector embeddings, and stored in a vector database. When a user submits a query, the system retrieves the most relevant contract sections and supplies them to an NVIDIA-hosted LLM through NVIDIA NIM Microservices.

The assistant can answer questions such as:

- “When does this contract expire?”
- “Does this contract contain an automatic renewal clause?”
- “What are the vendor’s service obligations?”

- “Are there any price escalation conditions?”

The extracted insights are displayed through a Contract Obligation Dashboard for efficient contract monitoring.

4. Methodology

- Collect vendor contracts, PO history, and procurement playbooks.
- Extract and preprocess text from uploaded documents.
- Generate embeddings using NVIDIA NeMo Retriever models.
- Store embeddings in a Milvus vector database.
- Retrieve relevant contract sections using semantic search.
- Use NVIDIA NIM-hosted LLMs to extract obligations, renewal dates, and escalation clauses.
- Present insights and alerts through an interactive dashboard.

5. Technology Stack

Component	Technology
Programming	Python
LLM Platform	NVIDIA NIM Microservices
Models	Llama 3 / NVIDIA Nemotron
RAG	NVIDIA AI Blueprint for RAG
Embeddings + Vector DB	NVIDIA NeMo Retriever + Milvus
Backend + Frontend	FastAPI + Streamlit
Document Processing	PyMuPDF, pdfplumber
Visualization	Plotly
Deployment	NVIDIA AI Workbench

6. Expected Outcomes

- Reduced manual contract review effort.
- Faster access to contract information through natural language queries.
- Improved visibility into vendor obligations and commitments.
- Automated identification of renewal dates and escalation clauses.
- Proactive alerts for expiring contracts and compliance risks.
- Enhanced procurement decision-making through AI-driven insights.

7. System Outputs & Business Impact

The proposed solution will generate a centralized **Contract Obligation Dashboard** that provides procurement teams with real-time visibility into critical contract information. The system will automatically extract and display:

- Contract renewal and expiration dates
- Vendor obligations and service commitments
- Payment terms and escalation clauses
- Compliance requirements and risk indicators
- Renewal alerts and contract status summaries

By transforming unstructured procurement contracts into actionable insights, the platform will reduce manual effort, improve contract governance, support faster procurement decisions, and help organizations proactively manage contractual risks and obligations.